

YUVRAJ SINGH

+91-7887249893 | mailatyuvraj Singh@gmail.com | [linkedin.com/in/yuvraj-singh009](https://www.linkedin.com/in/yuvraj-singh009) |

github.com/yuvraj-singh009 | leetcode.com/u/Yuvraj009

TECHNICAL SKILLS

- **Cloud & Infrastructure:** AWS (EC2, IAM, VPC), Terraform, Docker, Kubernetes
- **CI/CD & Monitoring:** GitHub Actions, Jenkins, Prometheus, Grafana, PromQL
- **Programming and Backend:** Java, Python, Bash, FastAPI, Flask, REST APIs
- **Databases & Tools:** MySQL, Supabase, Git, GitHub, Linux(Kali)
- **Core CS:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS

PROJECTS

Automated Cloud Application & Observability Stack

- Engineered a robust Python/FastAPI REST API backend, achieving 99.95% uptime and reducing average response times by 30% through custom Prometheus metrics.
- Provisioned and version-controlled AWS infrastructure (EC2) as code (IaC) using Terraform for repeatable, automated deployments.
- Reduced deployment time from approximately 15 minutes to under 2 minutes through CI/CD automation.
- Orchestrated a multi-container environment using Docker Compose, integrating application, Prometheus, and Grafana services with inter-container networking and persistent volumes, reducing manual setup effort by over 80%.
- Simulated real-world failure scenarios via chaos engineering (stress testing) and validated system resilience using PromQL-based Grafana dashboards.

Production Kubernetes Reliability & Observability Platform

- Containerized and deployed a FastAPI microservice on a Kubernetes (kind) cluster using Deployments, Services, ConfigMaps, and liveness probes to ensure high service availability and automatic recovery from failures. Implemented custom application observability using Prometheus client libraries, exposing 2 core metrics (request count, latency histogram) scraped every 10 seconds via a Kubernetes ServiceMonitor. Deployed Prometheus and Grafana using Helm, built real-time dashboards for request rate and p95 latency, and performed root cause analysis using application metrics during simulated workload spikes.

Flask Application on AWS using Terraform & Jenkins

- Engineered a fully automatic CI/CD pipeline to build and deploy a containerized 2-tier Flask + MySQL application on AWS EC2 using Jenkins, Docker, and Docker Compose, triggered on every Git push.
- Provisioned and managed AWS infrastructure (EC2, security groups, SSH key pairs) as code using Terraform, replacing manual console setup with a single reproducible terraform apply/destroy workflow.
- Automates server bootstrapping (Docker, Jenkins, Java) via EC2 user-data scripts, cutting environment setup time from a multi-step manual process to a fully hands-off deployment.

CERTIFICATIONS

- AWS Cloud Practitioner Certified | Issued : June 2026
- eJPTv2 Certified | Issued : Feb 2026
- The Bits and Bytes of Networking — Google (Coursera)
- Artificial Intelligence and Machine Learning -NPTEL

EXPERIENCE

- **Core Member –EV Club VIT Bhopal (Technical Team)**
 - Built and maintained technical solutions while supporting EV-focused events and collaborative projects.
- **Core Member – Hindi Club VIT Bhopal (Technical Team)**
 - Managed the club's technical infrastructure and collaborated on digital initiatives for events and community engagement.

EDUCATION

- **VIT Bhopal University** — B.Tech in Computer Science and Engineering, CGPA: 8.33 Aug23-27
- **Sant Atulanand Convent School** — Senior Secondary Education, 83.6%
- **Sant Atulanand Residential Academy** — Secondary Education, 90.8%